



ECNG 2005 Lab and Project Design III

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Semester II 2025 / 2026

1. GENERAL INFORMATION

Lab #:	CTRL-LAB1		
Name of the lab:	Parameter Estimation of a DC Motor Model		
Lab weighting:	7.5%	Estimated total study hours:	5 hrs
Delivery mode:	Lab		
Venue for the lab:	Control Systems Lab		
Lab dependencies:	The theoretical background to this lab is provided in ECNG2009		
Recommended prior knowledge and skills:	The prerequisite knowledge for this lab is detailed: 1. Use Laplace Transfer Functions 2. Determine the input/output response of Linear Time Invariant systems using the Laplace transfer function 3. Obtain the poles and zeros of LTI systems 4. Discuss why linear model representations are favoured for dynamic systems modelling 5. Use block diagrams to represent linear systems		

Course Staff	Position/Role	E-mail ✉	Phone ☎	Office	Office Hours
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2. LAB LEARNING OUTCOMES

Upon successful completion of the lab assignment, students will be able to:	Cognitive Level
1. Determine an accurate first order Linear Time Invariant model of a DC motor operated in armature control mode using two methods: a. First principles using manufacturer supplied data b. Input/Output testing	Analysis
2. Assess the accuracy and validity of the derived models	Evaluation

3. PRE-LAB

3.1. Prelab Exercise

1. Compare the specifications in Table 7: DCMCT Model Parameter Specifications in the Appendix of this lab manual to those in DC Motor Tutorial reading resource on myelearning. Are they the same?
2. The first row of Table 6 of this lab manual lists K and τ values determined in the solution of the DC Motor Tutorial. Under what conditions are these K and τ values valid?

3.2. Required Reading Resources

Katsuhiko Ogata, Prentice Hall 1997. *Modern Control Engineering* (4th Ed)

3.3. Other Resources

Course Web site on myelearning: <http://myelearning.sta.uwi.edu/login/index.php>



4. IN-LAB

Allotted completion time:	3 hours
Required lab Equipment:	1. The Quanser DCMCT rig 2. Power adapter for Quanser DCMCT rig 3. Serial / USB cable 4. Computer with serial/USB port and QICii software and JAVA engine installed.

4.1. Background

A servomechanism is a system used for controlling mechanical speed or position and consists of four major components:

1. **A servomotor** – This is a motor that allows precise control of its position and velocity. Servomotors may be either electric, pneumatic or hydraulic. The servomotor used in this lab is of the electric type and converts electrical energy to mechanical energy. The motion produced by this motor is rotational, but it should be noted that linear motors also exist.
2. **A controller and control amplifier** – These facilitate the automatic control of either speed or position of the servomotor.
3. **Velocity and position feedback sensors** – These measure the motion of the servomotor. These output measurements can be used to determine open loop responses or closed loop control.
4. **Gearbox or belt/pulley system (optional)** – These facilitate matching of the motor and load characteristics, just as a transformer is used in electric circuits.

Typical servomechanism applications include conveyor belts, consumer equipment drives (e.g. video and audio tapes, CDs, DVDs, hard-disc drives), aileron control in air craft, crane lifts, robotic manipulators etc.

4.2. Lab Equipment

This lab utilizes a DC Motor Control Trainer rig (DCMCT) manufactured by the company Quanser. Mounted on the DCMCT rig is a Maxon 18-Watt permanent magnet DC servomotor. This trainer rig allows the user/student to operate /control the Maxon servomotor via Personal Computer (PC) or external analog electronics. For this lab, we will use the QIC Processor Core under PC control to provide the excitation signals and measure and display motor variables. The DCMCT shown in Figure 1 consists of:

1. a potentiometer for speed and position sensing
2. a digital shaft encoder for greater precision speed and position sensing.
3. a *QIC Processor Core* driven by a PIC microcontroller for control and monitoring of the DCMCT from a PC connected to the DCMCT serial port.

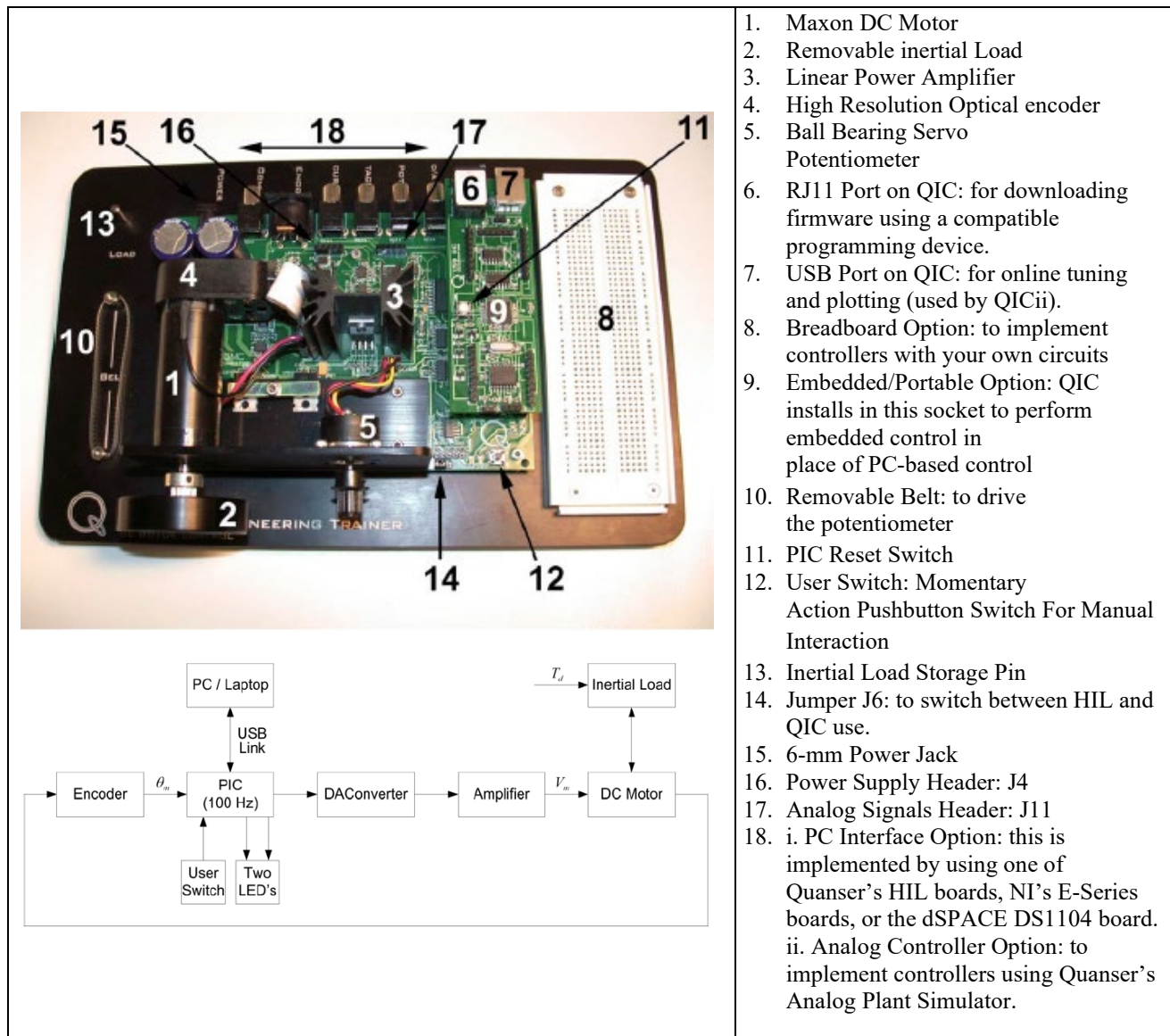


Figure 1: DCMCT General Layout, Component Nomenclature and simplified block diagram for this lab

CAUTION

It is extremely important that students do the following:

1. Note the ID number written on the USB Circuit Board of the DCMCT rig that you are using.
2. The DCMCT is an expensive piece of equipment which has exposed electronics. It is therefore important that extreme care be exercised when using the equipment.
 - a. DO NOT use food and drink at the workstation.
 - b. DO NOT allow metal jewellery to come into contact with exposed wiring.



4.3. QICii Modelling Module Startup

1. Ensure that the appropriate cable is connected to the USB B port on the Quanser DCMCT rig and USB A port on the PC.
2. Ensure that the power supply adapter is well connected to the DCMCT and the mains.
3. Start the QICii software installed on the PC by selecting **Start->Programs->Quanser->QICii USB->QICii USB**.

Note: A command window will appear before the QICii screen is launched.

4. Set the drop down menu in the upper left portion of the QICii screen to **Modelling**.
5. Press the Connect/Disconnect button  to connect.
6. Press the User Switch (refer to Figure 1).

Note: If successful, the removable inertial load should start to spin, and traces similar to those in Figure 1, should begin drawing in the display of the QICii software.

4.4. Description of QICii Modeling Module

In this lab, the Modelling module of the QICii software will be used to determine the speed-voltage characteristic and the open loop model of the DC motor. By selecting “Modelling” in the drop down box on the top left of the QICii window, the PIC on the board is then configured to excite the motor and measure its response. The Modelling Module user interface is shown in Figure 2 and the corresponding features of the Modelling Module user interface are listed and described in Table 1.

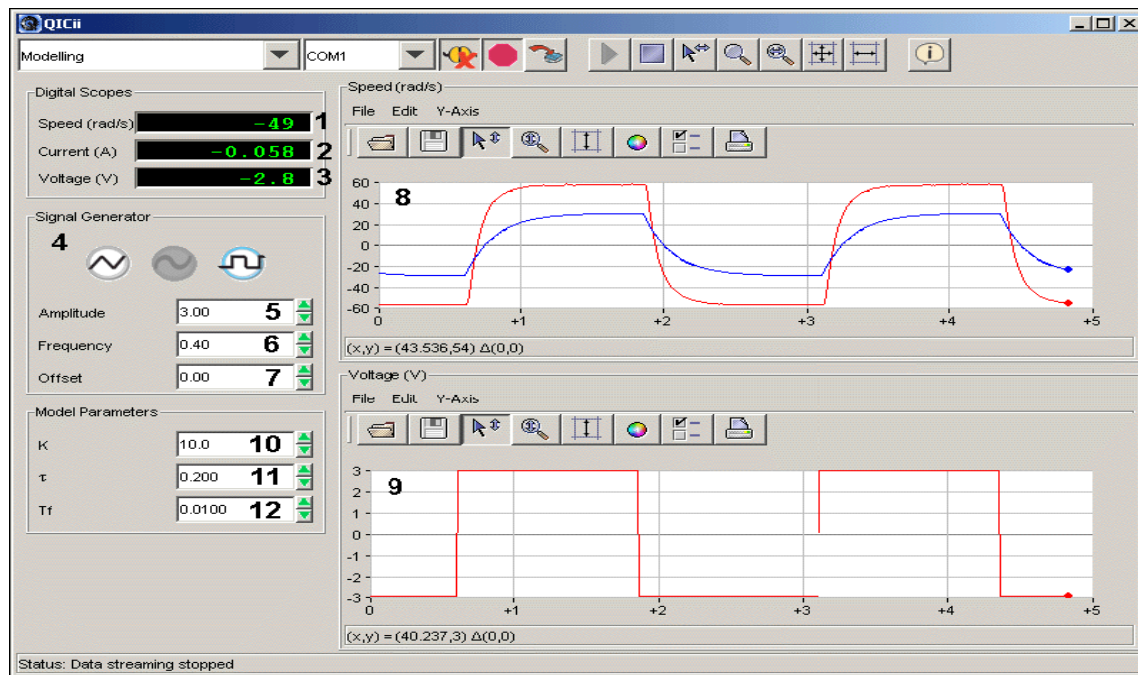


Figure 2: Modelling Module User Interface of the QICii Software



Table 1: Main features of Modelling Module User Interface (shown in Figure 2)

	Label	Parameter	Description	Units
1	Speed	Ω_m	Numeric Display of Motor Output Speed	rad/s
2	Current	I_m	Numeric Display of Motor Armature Current	A
3	Voltage	V_m	Numeric Display of Motor Input Voltage	V
4	Signal type		Type of Input Voltage signal supplied to motor (e.g. triangular, sine, square)	
5	Amplitude		Amplitude of Input Voltage signal supplied to motor	V
6	Frequency		Frequency of Input Voltage signal supplied to motor	Hz
7	Offset		Offset of Input Voltage signal supplied to motor	V
8	Speed	Ω_m	Graphs of Actual (red) and Simulated (blue) Motor Speeds vs Time	rad/s
9	Voltage	V_m	Graph of Input Voltage signal supplied to motor (red) vs Time	V
10	K	K	Input Box for Motor Model Steady-State Gain	rad/(V.s)
11	τ	τ	Input Box for Motor Model Time Constant	s
12	Tf	T_f	Time Constant of Filter for Measured Signal	s

NOTE:

- The PIC does not “close the loop” to implement feedback control. Instead, it performs the roles of a signal generator and an oscilloscope.
- Signals applied to the motor rig are not continuous analog signals but are digitized at a sample rate of 100 Hz. Since the sampling rate is much faster than the maximum motor speed, this should not pose a problem.
- Motor speed is **estimated** from the derivative of its position as measured by the encoder. As derivative elements amplify high frequency noise, the actual speed is obtained by first filtering the position signal $\left(\frac{1}{T_{fs}+1}\right)$. This results in the following filter: $\omega_m(s) = \frac{s}{T_{fs}+1} \theta_m(s)$ where θ_m is the position of the motor shaft measured by the encoder.

4.5. Lab Tips

- The lab requires that you change the input voltage several times. This can be done either by
 - using the green **up/down arrows** on your QICii interface or
 - Entering the values using the **keyboard** of your PC.

Note: If using the keyboard, the **ENTER** key **MUST** be pressed after entering the new value.

- In the event of disconnection, perform the following:
 - Press the User Switch.
 - Press the Connect/Disconnect button to reconnect.



4.6. Velocity-Voltage Characteristics

In this section, the velocity-voltage characteristic of the motor will be experimentally determined. The Offset of the signal generator voltage will be used to input DC voltages to the motor. This will allow us to monitor the steady state speed and direction of movement for each voltage setting. *The current value in the numerical display remains relatively constant despite some fluctuation.*

4.6.1. Increasing Positive Offset

1. Set the Amplitude and Offset of the Signal Generator to zero. *Note: Since a DC voltage is to be applied, the AC portion is set to zero and only the DC offset voltage of the function generator will be varied. Since there is AC, the frequency setting will become irrelevant.*
2. Gradually increase the Offset in steps of 0.01V until the motor starts to move continuously. *This voltage is the **Positive Breakaway** voltage, V_{pb} .*
3. Record the **Positive Breakaway** voltage, V_{pb} in the first row of Table 3 and the corresponding steady state velocity in the “Increasing” section of Table 3. *The steady state value for speed can be obtained from the numeric display in the top left corner of your QICii window.*
4. Increase the offset to the **next multiple of 0.1V that is higher than V_{pb}** . Cross out any offset values in Table 3 between V_{pb} and this value.
5. Record the steady state velocity for this value in the “Increasing” section of Table 3.
6. Increase the positive offset to determine the steady state velocities for the remaining positive offsets in Table 3. Record these steady state velocities in the “Increasing” section of Table 3.

4.6.2. Decreasing Positive Offset

7. Using the same Offset values as in 4.6.1, record the steady state velocities for each offset, starting from $V_m = 5V$ and then decreasing. Record your results in the “Decreasing” section of Table 3.

4.6.3. Increasing Negative Offset

8. Repeat 4.6.1 but for negative voltages ($-5V \leq V_m \leq 0V$), by recording the **Negative Breakaway voltage** V_{nb} in the first row of Table 4, crossing out the unused negative offset values and recording the steady state velocities in the “Increasing” section of Table 4.

Note that the inertial load is now spinning in the opposite direction, so the velocities Table 4 in should be recorded as a negative values.

4.6.4. Decreasing Negative Offset

9. Using the same negative offset values as in 4.6.3, record the steady state velocity for each offset, starting from $V_m = -5V$ and then decreasing in magnitude. Record your results in the “Decreasing” section of Table 4.



4.7. Dynamic Models: Experimental Determination of System Dynamics

4.7.1. Step Response Tests

A linear model of a system can be determined experimentally by observing how the system reacts to different inputs. In this section, an approximate first order model for the motor will be determined using step response test. The first order model of the system is of the form $\Omega_s(s) = \frac{K}{\tau s + 1} V_m(s)$ where Ω_s is the motor speed and V_m is the input voltage. The step response test rapidly changes the input voltage from one voltage to another, thus taking the system from one steady state to another. Thus the input changes by Δu and the corresponding changes in the steady state output is Δy . An estimate of the steady-state gain K can then be determined using the formula $K = \frac{\Delta y}{\Delta u}$. The time constant τ can also be interpolated from the step response graph as it is the time taken for the output to reach approximately 63.2% of Δy .

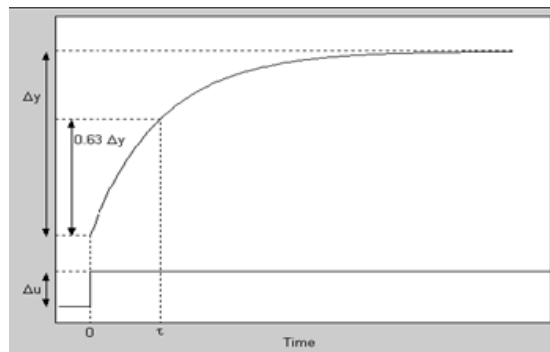


Figure 3: Graph of Step Response Test Showing the Input and Output

Examine Figure 4 in the appendix. This figure describes some important QICii toolbar buttons. You will use these buttons in this section to start and stop the plots, to take measurements from the plots, or to determine point coordinates. Hover over each toolbar button in the QICii window to obtain a description of their purpose.

Procedure

1. Set the Amplitude and Offset of the Signal Generator to zero.
2. Set the offset to 3V and wait until the system settles.
3. Increase the offset from 3V to 5V ($\Delta u = 2V$). Stop the display when the transition point has reached almost to the left half of the screen (similar to Figure 4). The response should have settled by then.
4. Use this plot to determine the Δy and the time constant, τ of the red graph (the actual response of DCMCT). Record your results in Table 5. *Tip: Use the Ctrl feature of the magnifier icons.*
5. Calculate the steady state gain, K (using the formula) and record your result in Table 5.
6. Repeat steps 2 to 5 for $\Delta u = -2V$.
7. Set the offset to -3V and repeat for $\Delta u = \pm 2V$ as above. Record your results in Table 5.
8. Complete Table 5.



4.7.2. Validation using Pulse inputs

In this section, the motor will be pulsed while a 1st order simulation model is adjusted to match the observed DCMCT responses. In the previous section, approximate values for the steady state gain K and time constant τ were determined. The output response of this approximate first model with parameters K and τ , however, must be compared to the actual response of the DCMCT rig in order to determine whether this model is a true reflection of the rig's behaviour. This process is called *validation*. Since the QICii software is capable of simultaneously displaying the output response of a 1st order simulation model and the output response of the DCMCT rig, we can determine whether our calculated model satisfactorily represents the DCMCT rig. The output of this simulated model will be displayed in blue whereas the actual output of the DCMCT rig will be seen in red in the QICii window.

In addition, if we wish to improve the calculated model, we can adjust its parameter values of the simulated model until the actual output of the DCMCT rig (seen in red) is approximately the same as the simulation output (seen in blue). Since they are driven by the same signal, by changing the structure and parameters of the reference model until a reasonable fit between the responses of the two models is obtained, we can obtain the best possible model. This process is called *model fitting*. This “fitted” simulation model can therefore be used to estimate the model parameters for the DCMCT rig. Thus the QICii software can determine the best-fit 1st order model for the motor through *model fitting and validation*.

Procedure

1. Apply a square wave input voltage (with dc offset) from the signal generator to the open-loop system by setting the QICii module parameters as described in Table 2. This applies a series of Step input voltages to the motor. As in the step response tests, the motor is biased at 3V input and then subjected to a $\pm 2V$ step input variation. The period of 2.5s is so long that the system reaches steady-state with each step.

Table 2: Module Parameters for the Step Response Test

Signal Type	Amplitude /V	Frequency / Hz	Offset / V
Square Wave	2	0.2	3

1. Set K and τ of the simulated model to zero.
2. Change the simulation parameters K and τ to the values that were previously determined in the step response test. Print screen your result.
3. Change the simulation parameters K and τ to the values calculated in the tutorial (refer to Prelab). Print screen your result.
4. Adjust the K and τ parameters of the model so that the simulated and actual outputs are as close as possible. Record the values of K and τ in Table 6 and print screen your result.
5. Complete Table 6.



4.8. In-Lab Results

Table 3: Velocity vs Positive Offset

Positive Offset /V (V _{pb})	ω_m / rad s ⁻¹	
	<i>Increasing</i>	<i>Decreasing</i>
0.1		
0.2		
0.3		
0.4		
0.5		
0.6		
0.7		
0.8		
0.9		
1.0		
1.5		
2.0		
2.5		
3.0		
3.5		
4.0		
4.5		
5.0		



Table 4: Velocity vs Negative Offset

Negative Offset /V (V _{nb})	ω_m / rad s ⁻¹	
	<i>Increasing</i>	<i>Decreasing</i>
-0.1		
-0.2		
-0.3		
-0.4		
-0.5		
-0.6		
-0.7		
-0.8		
-0.9		
-1.0		
-1.5		
-2.0		
-2.5		
-3.0		
-3.5		
-4.0		
-4.5		
-5.0		



Table 5: Step Response Parameters

Initial Input	Final Input Voltage/ V	Δu /V	Δy / rad s ⁻¹	$0.632 \Delta y$ / rad s ⁻¹	K /rad s ⁻¹ V ⁻¹	τ /s
3		2				
3		-2				
-3		2				
-3		-2				
Average						

Table 6: Summary of K and τ values

	K /rad s ⁻¹ V ⁻¹	τ /s
From Tutorial (See Prelab)	19.86	0.093
From Step Response Tests		
From “Model Fitting”		

Proceed to post-lab exercise.



5. POST-LAB

Due Date:	Friday 13th March 2026 @ 11PM
Submission procedure:	• Acknowledge the online Submission statement concerning plagiarism. • Upload pdf to the designated Post-Lab submission Link on myelearning.
Deliverables:	Post-Lab Report containing • Lab Coversheet (NB The template is available on myelearning), • <i>Typewritten</i> responses to the assignment questions <i>consistent with the Postlab question numbering</i>, • Appropriately labelled <i>In-Lab results</i> with all relevant information included.

5.1. Assignment

Include in the Appendix, the results obtained from the lab: Table 3 [12 marks], Table 4 [12 marks], Table 5 [15 marks], Table 6 [3 marks].

1. State the objectives of the lab [6 marks]

5.1.1. Velocity-voltage Characteristic

2. Produce the Velocity-voltage characteristic for the motor over the test range, by plotting the data from Table 3 and Table 4 on a single graph. [10 marks].
3. Determine the average value of DC gain, K , of the system from the slope of the characteristic [3 marks].
4. Discuss the shape of the curve around $V_m = 0$ [3 marks].
5. Discuss the shape of the curves for the negative and positive V_m . [3 marks].
6. Explain whether or not a linear model is a valid for this motor [4 marks].

5.1.2. Dynamic Models: Experimental Parameter Estimation of System Dynamics

7. Discuss the validity of the First order LTI model (refer to Section 4.7) used to represent the DC motor [8 marks].
8. Discuss whether a good fit between the estimated response (i.e. plot obtained by setting K and τ to the average values obtained from the step response test, Table 5) and actual responses were obtained. [3 marks].
9. Discuss whether there was a good fit between the estimated response (i.e. plot obtained after applying the K and τ values previously determined in the tutorial/ Prelab) and the actual responses was obtained [5 marks].



10. Compare and discuss the K and τ values in Table 6 and their resulting graphs. Explain how the different techniques used to obtain the values caused variation in the results and discuss how well each of the techniques used fits this model to the data **[9 marks]**.
11. State which set of results should be chosen as the model parameters for the DC motor and why? **[4 marks]**.

Total: 100 marks



6. APPENDICES

6.1. QICii Toolbar buttons



Figure 4: QICii Toolbar buttons



6.2. System Parameters according to Manufacturer

Table 7: DCMCT Model Parameter Specifications

Symbol	Description	Value	Unit
k_t	Motor Torque Constant	0.0502	Nm/A
R_a	Motor Armature Resistance	10.6	Ω
L_a	Motor Armature Inductance	0.82	mH
J_m	Moment Of Inertia Of Motor Rotor	1.16E(-6)	Kg.m ²
τ_m	Motor Mechanical Time Constant (unloaded)	0.0049	s
	Motor Mechanical Time Constant (loaded)	0.093	s
M_i	Inertial Load Disc Mass	0.068	Kg
r_i	Inertial Load Disc Radius	0.0248	M
k_e	Back EMF Constant	0.0502	V/(rad/s)
	Motor maximum continuous torque	0.035	Nm
	Motor power rating	18	W
	Linear Amplifier		
$V_{\max}$	Linear Amplifier Maximum Output Voltage	15	V
	Linear Amplifier Maximum Output Current	1.5	A
	Linear Amplifier Maximum Output Power	22	W
	Linear Amplifier Maximum Dissipated Power	8	W
	Linear Amplifier Gain	3	V/V

**End of CTRL-LAB1 Determination of the
DC Motor Model**